

Control-Plane Reference — FSMs & Flag Bits

One channel, DDR5. The scheduling pressure = the flags + FSM states (not the timing consts). Domains: mc=mc_clk, ctrl=DFI control-clk, dev=device. Widths parametric.

1 FSMs running

FSM	count	states	outputs (per state)	read by
BANK_FSM	×N_BANKS	IDLE ACTING OPEN PREING REFING RFMING	state[4]; row open only in OPEN; cas_complete=(IDLE); bank_busy!=(IDLE); ACTING->OPEN via tRCD, PREING/REFING/RFMING->IDLE via timers	ELIG (need/can_cas), arb, refresh/RFM (cas_complete), maint
REFRESH_FSM	×1	IDLE PICK DRAIN ISSUE REFING release	PICK: stall_acts[8], ba; DRAIN: AP-drain; ISSUE: cmd=REF, gate_rfc[8]; release: debt<=0	scheduler (o_ref_rdy, ref_force), banks (stall_acts)
RFM_FSM	×1	IDLE PICK DRAIN ISSUE RFMING release	like REFRESH but cmd=RFM, gate_rfm, RAA[8]=RAAIMT; trigger RAA	scheduler (rfm_rdy, rfm_force), banks
MAINT_PRIO	comb	-	picks cmd_type among pending ref/rfm/zq/mrw/mrr; init separate	phaser / DFI encode (cmd_type)
INIT_FSM	×1	RESET CKE MRW_TRAIN ZQCL PHY_HS DONE	dfi_reset_n; drives DFI during init; DONE sets init_done (one-way)	DFI_MUX (sel), all blocks (gate until init_done)
FREQ_FSM	×1	IDLE QUIESCE SRE FREQ_HS RELOCK REPROG SRX RESUME	drives ctrlupd/freq_ratio; reloads timing/tphy_*/MR; updates gear at RESUME	PHY, cfg regs, scheduler (gear)
POWER_FSM	×1 (per rank)	ACTIVE ACT_PD PRE_PD SREF	dfi_cke, dfi_dram_clk_disable, dfi_lp_*; enters on idle	DFI, banks (state)
ZQ_FSM	×1	IDLE ZQCAL ZQLAT	zq_rdy, gate_zq; DQ cal busy	scheduler (zq_rdy)
MRW_FSM	×1	IDLE ISSUE MRD MOD	mrw_rdy, gate_mr (tMRD), setting live (tMOD)	scheduler (mrw_rdy)
MRR / MR_POLL	×1	IDLE ISSUE WAIT CAP	mrr_rdy; captures MR4 -> rr_shift (temp); RD_CAP sideband	scheduler, refresh (rr_shift)
WLR_FSM	×N_RANKS	(batch_dir + streak/age)	batch_dir, wlr_select; adaptive batch len	ingress admit, L3 dir-bias
WR_ROB	×ROB	ADMIT FILL ISSUE RETIRE	{vld,N,gen_idx,done_cnt}; wr_pack; retire at done_cnt==N	write path, completion

2 Flag / status bits to the scheduler

Per-bank legality (the can_* gates + demand)

flag/status	width	from	to	meaning / asserted when
can_act / can_cas / can_pre	1 ea ×N_BANKS	TREG cmp + STATUS (can_cas needs OPEN)	ELIG / CMD_ARB	cmd electrically legal for this bank now (GC>=next_* & state)
need_act / need_cas / need_pre	1 ea ×N_BANKS	row_comparator (cell)	ELIG	what the resident packet wants (hit->cas, miss->pre/act)
is_hit	1 ×N_BANKS	row_comparator	CAS_W weight	open_row == pkt row
row_valid / pre_pending	1 ea ×N_BANKS	BANK_STATUS	ELIG	admit window (OPEN & !pre_pending); PRE-propose drops it
bank_go	1 ×N_BANKS	ELIG (need & can)	BGB_ARB	bank has an eligible command

Cross-scope legality

flag/status	width	from	to	meaning / asserted when
can_act_bgs / can_cas_bgs	1 ea ×N_BG	BG scope cmp	BGB/BG_ARB	same-BG spacing met (tRRD_L / tCCD_L(_WR))
can_act_bgd / can_act_rank	1 ea ×N_RANKS	rank scope (tRRD_S, faw)	RANK/L3	cross-BG ACT + 4-ACT window ok
can_wr / can_rd (turnaround)	1 ea ×N_RANKS	rank turnaround	L3 CAS lane	R<->W flip gap met (tWTR/tRTW)
can_cas_dq	1	global dqFree ctr	L3 CAS lane	DQ bus free (BL/2); cross-rank tRTRS folded

Maintenance / refresh (ME -> scheduler)

flag/status	width	from	to	meaning / asserted when
o_ref_rdy	1	REFRESH (debt)	scheduler	a refresh is owed (opportunistic if bank idle)
ref_force	1 ×ba	REFRESH (&debt==7)	scheduler	forced refresh (drain + go)
rfm_rdy / rfm_force	1 ea ×ba	RFM (RAA>=IMT / >=MMT)	scheduler	rowhammer owed / hard (blocks next ACT)
zq_rdy / mrw_rdy / mrr_rdy	1 ea	ZQ / MRW / MRR	scheduler	that maint op ready to issue
stall_acts	1 ×N_BANKS	maint (OR of needs)	bank FSM	quiesce: freeze admit + block new ACT -> drain
cas_complete	1 ×N_BANKS	BANK_FSM (state==IDLE)	maint (drain done)	bank drained + precharged, ready for maint
gate_rfc / gate_rfm / gate_zq / gate_mr	1 ea (banks/rank)	maint FSMs	can_* mask	block issue during REFING/RFMING/ZQ/tMRD
init_done	1	INIT_FSM	DFI_MUX + all	boot complete; flips CA driver init->scheduler

Grants + DFI status

flag/status	width	from	to	meaning / asserted when
grant_oh (cas/act/pre)	N_BANKS ea	L3 arb	cells	per-class winner one-hot; pop-on-CAS; commits FSM
cmd_type	5	arb / MAINT_PRIO	DFI encode	which command issues (demand or maint)
dfi_init_complete	1 (in)	PHY	INIT/FREQ FSM	PHY init/freq-change ack
dfi_phyupd_req / dfi_ctrlupd_ack	1 ea (in)	PHY	UPDATE ctrl	PHY-requested update / ctrl-update grant
dfi_alert_n / dfi_error	1 ea (in)	PHY	ALERT/ERR mon	CA-parity/write-CRC alert; PHY error

Key seams: (1) legality folds into `can_*` at ELIG so the arb only selects (never rejects). (2) Maint uses ONE shared seam: `stall_acts` (drain req) + `cas_complete` (drained) + `cmd_type` (which op), MAINT_PRIO picks. (3) INIT is the only hard override (DFI_MUX). (4) `grant_oh` is per-class (cas/act/pre), pop-on-CAS. All control bits live in `mc_clk`; DFI status ins (`init_complete/phyupd/alert/error`) are control-clk, sampled in `mc_clk`.